Cohort Batch Correction in SSBMF: Literature Summary and Implementation Outline

Andrew Walther

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The Problem

Supervised Sparse Bayesian Matrix Factorization (SSBMF) jointly fits a low-rank genomics factor model and a Cox proportional hazards survival model. The genomics reconstruction is $Y = LF' + E$ and the survival linear predictor is $\eta = L\beta$. When the model is trained on a single cohort — TCGA or CPTAC individually — prognostic factors are recovered and external C-indices reach 0.55–0.67.

When training data from TCGA (RNA-seq) and CPTAC (proteomics) are pooled, the dominant axis of variation in Y is platform technology, not biology. This platform factor captures the first principal component, and the survival signal is too weak to compete: the CAVI algorithm converges with $\hat{\beta} \approx 0$, producing a prognostic score of no clinical value. Per-platform z-score normalization mitigates but does not eliminate the collapse, and discards cohort-specific information in the process. A principled solution requires modeling platform effects *within* the inference procedure, not as a preprocessing step.

Literature Context

Mixed models for batch effects. The linear mixed model (LMM) is the standard framework for separating signal from structured nuisance variation. In the frequentist tradition, restricted maximum likelihood (REML) estimates variance components after marginalizing over fixed effects (Chen and Dunson 2003). Bayesian formulations place priors directly on variance components; Gelman (2006) recommended half-Cauchy priors as a weakly informative default that avoids the well-known pathologies of inverse-gamma priors near zero. George and McCulloch (1993) introduced spike-and-slab selection for random effects; Carvalho, Polson, and Scott (2010) and Piironen and Vehtari (2017) developed the horseshoe and regularized horseshoe as continuous relaxations with better numerical behavior. For our setting — $C = 2$ to 7 cohorts with $n_c \approx$ 50–300 patients each — the relevant message from this literature is that borrowing strength across groups via a shared variance hyperprior is most beneficial when $C \gg 1$ and group sample sizes are small. With only two cohorts and ≥ 100 patients per cohort, independent Normal priors on the cohort offset parameters are well-identified and require no hyperprior.

EBNM and EBMF — the SSBMF methodological ancestor. Stephens (2017) introduced the Empirical Bayes Normal Means (EBNM) framework, in which the prior on a set of unknowns is estimated from the marginal likelihood rather than fixed in advance. Wang and Stephens (2021) extended this to Empirical Bayes Matrix Factorization (EBMF): the model $Y = LF' + E$ is fit by coordinate ascent variational inference in which each factor update reduces to an EBNM problem, automatically selecting the appropriate amount of sparsity. The `ebnm` R package (Willwerscheid, Carbonetto, and Stephens 2025) provides the computational backend. SSBMF is a supervised extension of EBMF that adds a Cox partial likelihood on $\eta = L\beta$; all existing update functions call EBNM internally. The proposed cohort extension fits naturally within this infrastructure — the cohort offset factors have a simple Normal prior, giving a closed-form Gaussian posterior that requires no EBNM optimization.

Genomics batch-effect correction methods. ComBat (Johnson, Li, and Rabinovic 2007) corrects additive batch shifts gene-by-gene via empirical Bayes shrinkage and is the de facto standard preprocessing method for bulk RNA-seq; ComBat-seq (Zhang, Parmigiani, and Johnson 2020) extends this to count data. SVA (Leek and Storey 2007) estimates latent confounders when batch labels are unknown; RUV (Risso et al. 2014) uses control genes to identify technical factors. All four are preprocessing steps that transform Y before downstream analysis, decoupling batch correction from the survival inference objective.

Among multi-omics factor analysis methods, MOFA (Argelaguet et al. 2018) is the architecturally closest to SSBMF: it learns a shared low-dimensional representation across omics layers via variational Bayes with ARD sparsity priors. MOFA+ (Argelaguet et al. 2020) extends this to multiple sample groups with separate loading matrices per batch. Harmony (Korsunsky et al. 2019) and scVI (Lopez et al. 2018) perform batch integration in embedding space. DIABLO (Singh et al. 2019) performs supervised multi-omics integration but targets categorical phenotypes and cannot constrain cohort effects out of a continuous prediction score.

Why existing methods are insufficient. Every method above either decouples batch correction from the survival objective (ComBat, SVA, RUV), targets an unsupervised objective (MOFA, MOFA+, Harmony, scVI), or cannot exclude cohort effects from prediction (DIABLO). None jointly optimizes a survival objective while estimating platform-specific offsets within the same iteration. MOFA+ is the closest relative but has no survival component and no $\beta_{\text{cohort}} \equiv 0$ constraint.

Proposed Solution

The proposed extension appends C cohort indicator columns to L, giving $Y = L_{\text{global}}F'_{\text{global}} + L_{\text{cohort}}F'_{\text{cohort}} + E$ where L_{cohort} is a fixed binary indicator matrix and F_{cohort} (estimated) absorbs platform-specific mean expression offsets. The survival linear predictor remains $\eta = L_{\text{global}}\beta_{\text{global}}$ with $\beta_{\text{cohort}} \equiv 0$ fixed. Three design principles motivate this formulation:

1. **Closed-form updates.** Because L_{cohort} is fixed and the Normal prior is conjugate to the Normal genomics likelihood, the posterior $q(F_{\text{cohort},cj})$ has an explicit Gaussian form — no EBNM optimization is required. The computational overhead is $O(np)$ per iteration, roughly 40% of a single global factor update for $C = 2, K = 5$.
2. **Sign symmetry.** A cohort may have systematically higher or lower expression than average for any gene. The one-sided sparsity priors (point-Normal, point-exponential) used for global factors are inappropriate here; a symmetric Normal prior centered at zero is correct, particularly because Y is column-centered before fitting.
3. **Protected prognostic score.** Fixing $\beta_{\text{cohort}} = 0$ is not a computational shortcut — it is a deliberate constraint that prevents the Cox likelihood from exploiting cohort-level survival differences (which may reflect censoring patterns or follow-up length rather than biology). The resulting score depends only on global biological factors and generalizes to patients from cohorts not seen at training time.

Implementation Outline

The extension requires five targeted modifications to the existing modular code pipeline, organized as five sessions:

Session	Deliverable	Gate
A	LaTeX derivation (`derivations/qF_cohort/`)	Advisor math review

Session	Deliverable	Gate
B	New module `code/update_F_cohort.R` + tests + demo	15/15 new tests pass
C	Modifications to `compute_R_k`, `update_F_all`, `compute_var_term`, `compute_elbo.R`, `fit_modular.R` + integration tests	~218 tests pass; `cohort_id=NULL` reproduces current results exactly
D	Synthetic validation (`results/cohort_lmm_sim/`) — known ground truth, three-model comparison	$\hat{\beta}$ recovery $r > 0.9$; F_{cohort} recovery $r > 0.8$; C-index $\geq$ baseline
E	Real TCGA + CPTAC benchmark + results report	$\hat{\beta}$ non-collapse; external C-index ≥ 0.60 on ≥ 2 of 4 held-out cohorts

The existing 193 passing tests are unaffected by design — backward compatibility is verified in Session C before any validation is run. The full three-part planning document with mathematical derivations, complete R pseudocode, and detailed acceptance criteria is in `cohort_lmm_review_plan.pdf`.

Full bibliography and mathematical derivations in `cohort_lmm_review_plan.pdf`.